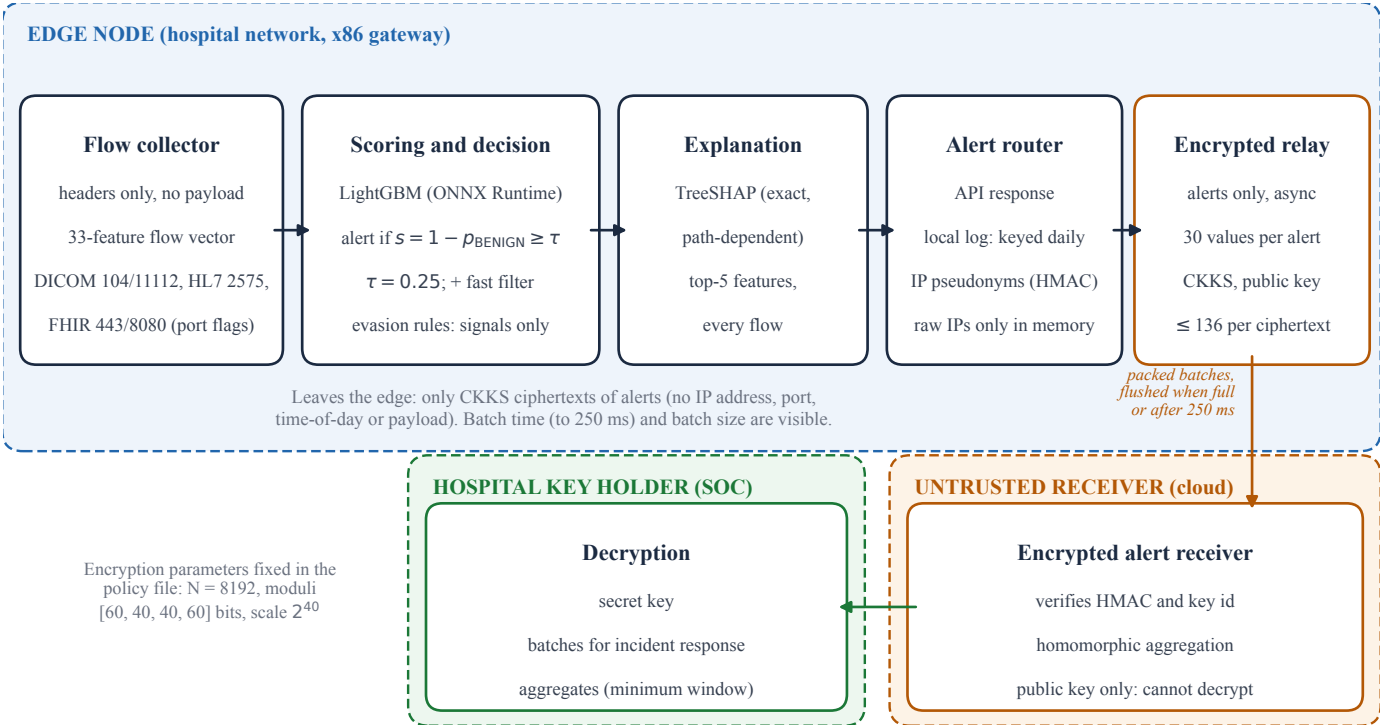

EDGE NODE (hospital network, x86 gateway)
Flow collector
headers only, no payload
33-feature flow vector
DICOM 104/11112, HL7 2575,
FHIR 443/8080 (port flags)
Scoring and decision
LightGBM (ONNX Runtime)
alert if $s = 1 - p_{\text{BENIGN}} \geq \tau$
$\tau = 0.25$; + fast filter
evasion rules: signals only
Explanation
TreeSHAP (exact,
path-dependent)
top-5 features,
every flow
Alert router
API response
local log: keyed daily
IP pseudonyms (HMAC)
raw IPs only in memory
Encrypted relay
alerts only, async
30 values per alert
CKKS, public key
≤ 136 per ciphertext
packed batches,
flushed when full
or after 250 ms
Leaves the edge: only CKKS ciphertexts of alerts (no IP address, port,
time-of-day or payload). Batch time (to 250 ms) and batch size are visible.
Encryption parameters fixed in the
policy file: N = 8192, moduli
[60, 40, 40, 60] bits, scale 2^{40}
HOSPITAL KEY HOLDER (SOC)
Decryption
secret key
batches for incident response
aggregates (minimum window)
UNTRUSTED RECEIVER (cloud)
Encrypted alert receiver
verifies HMAC and key id
homomorphic aggregation
public key only: cannot decrypt